

“GreenSight: Smart Vegetation Monitoring Using AI & LiDAR Technology”

PRAJWALA P, YASHRAJ NAIK, ABHISHEK B

Corresponding Author Email ID: prajwala.28p@gmail.com

Dayananda Sagar College of Engineering, Bengaluru

SDG Goals : LIFE ON LAND.

Hackathon Problem Statement: “Traditional vegetation classification and manual system is slow. Hence an AI–LiDAR integration Module is needed for accurate, real-time species, biomass, and health analysis”.

Abstract (Maximum 700 Words)

Forests and agricultural lands are critical for biodiversity, climate regulation, and food security, yet their monitoring remains challenging when performed using traditional manual methods. *GreenSight* introduces a smart vegetation management system that combines LiDAR sensing and artificial intelligence to deliver precise, scalable, and efficient analysis. Using UAVs and autonomous platforms, LiDAR data and spectral imagery are captured to generate high-resolution 3D models of canopies and crops. Deep learning algorithms, particularly convolutional neural networks (CNNs), enable accurate species identification, biomass estimation, and vegetation health monitoring.

Important forest parameters such as Diameter at Breast Height (DBH), tree height, and canopy density can be estimated with high accuracy. Sensor fusion improves classification reliability, while edge-based AI supports real-time decision-making for disease detection, yield estimation, and sustainable resource planning. By automating data collection and analysis, *GreenSight* significantly reduces survey time and operational cost while improving ecological insight. The system benefits both forestry and agriculture, offering an innovative approach to precision forestry, smart farming, and biodiversity conservation through AI-driven LiDAR technologies.

Keywords: , LiDAR, Precision Forestry, 3D Point Cloud .